Midterm 2 Formula Sheet April 22, 2026.
V0.1 AAG

1. Generalized Reactor Continuity & Energy Equations

1.1 Generalized Species Continuity Equation (Tubular Reactor)

Name of the equation: Generalized Species Continuity Equation (Tubular Reactor)

The equation:

$$u_z(r)\frac{\partial C_j}{\partial z} = D_{e,z}(r)\frac{\partial^2 C_j}{\partial z^2} + \ldots$$
$$\ldots\frac{1}{r}\frac{\partial}{\partial r}\left(r\, D_{e,r}(r)\frac{\partial C_j}{\partial r}\right) + R_j$$

What the equation represents/how it's used: The steady-state mass balance on species j in a tubular reactor. It accounts for axial convection (left side), axial and radial dispersion, and reaction.

Explain each term and the units of each term:

- $u_z(r)$: axial fluid velocity at radial position r — units: m/s or cm/s
- C_j: molar concentration of species j — units: mol/m^3 or mol/L
- z: axial coordinate along reactor length — units: m
- $D_{e,z}(r)$: effective (dispersion) coefficient in the axial direction — units: m^2/s
- r: radial coordinate — units: m
- $D_{e,r}(r)$: effective (dispersion) coefficient in the radial direction — units: m^2/s
- R_j: volumetric reaction rate of species j (positive for formation, negative for consumption) — units: mol/(m$^3 \cdot$ s)

1.2 Generalized Energy Balance (Tubular Reactor)

Name of the equation: Generalized Energy Balance (Tubular Reactor)

The equation:

$$\sum_j M_j C_{p,j} C_j\, u_z(r)\frac{\partial T}{\partial z} = \ldots$$
$$\ldots\lambda_{e,z}(r)\frac{\partial^2 T}{\partial z^2} + \frac{1}{r}\frac{\partial}{\partial r}\left(r\,\lambda_{e,r}(r)\frac{\partial T}{\partial r}\right) + \sum_i(-\Delta H_i)R_i$$

What the equation represents/how it's used: The steady-state thermal energy balance in a tubular reactor. The left side is heat carried by convection; the right side has axial/radial conduction (dispersion) and heat generated (or absorbed) by reactions.

Explain each term and the units of each term:

- M_j: molecular weight of species j — units: kg/mol
- $C_{p,j}$: specific heat capacity of species j — units: J/(kg $\cdot$ K)
- T: temperature — units: K
- $\lambda_{e,z}(r)$, $\lambda_{e,r}(r)$: effective thermal conductivity in axial and radial directions — units: W/(m $\cdot$ K)
- ΔH_i: enthalpy of reaction i — units: J/mol
- R_i: rate of reaction i — units: mol/(m$^3 \cdot$ s)

1.3 Danckwerts Boundary Conditions

Name of the equation: Danckwerts Boundary Conditions

The equation:

At inlet ($z = 0$):

$$u_z(r)\, C_{j,0} = u_z(r)\, C_j(0,r) - D_{e,z}(r)\frac{\partial C_j(0,r)}{\partial z}$$

At outlet ($z = L$):

$$\frac{\partial C_j}{\partial z}(z = L) = 0$$

What the equation represents/how it's used: The physically correct boundary conditions for a reactor with dispersion. At the inlet, they account for the fact that diffusive flux can carry material back upstream (back-mixing). At the outlet, they enforce no concentration gradient (fully developed flow exits).

Explain each term and the units of each term:

- $C_{j,0}$: feed concentration of species j — units: mol/m^3
- L: reactor length — units: m
- All other terms defined above.

1.4 Laminar Velocity Profile (Hagen–Poiseuille)

Name of the equation: Laminar Velocity Profile (Hagen–Poiseuille)

The equation:

$$u_z(r) = 2\bar{u}\left(1 - \left(\frac{r}{R_t}\right)^2\right)$$

What the equation represents/how it's used: The parabolic, fully-developed laminar velocity profile inside a tube. The velocity is maximum at the center ($r = 0$) and zero at the wall ($r = R_t$).

Explain each term and the units of each term:

- $\bar{u}$: mean (cross-sectionally averaged) axial velocity — units: m/s
- r: radial position — units: m
- R_t: tube radius — units: m

2. Residence Time Distribution (RTD)

2.1 Exit Age Distribution Function $E(t)$

Name of the equation: Exit Age Distribution Function $E(t)$

The equation:

$$E(t) = \frac{C(t)}{\int_0^\infty C(t)\,dt} = \frac{C(t)}{M_0}$$

What the equation represents/how it's used: The probability density function of residence times. $E(t)\,dt$ gives the fraction of fluid that spent between t and $t + dt$ in the reactor. Measured from a tracer pulse experiment.

Explain each term and the units of each term:

- $C(t)$: tracer concentration measured at reactor outlet as a function of time — units: mol/m^3
- $M_0 = \int_0^\infty C(t)\,dt$: zeroth moment (total area under the tracer curve) — units: mol $\cdot$ s/m^3

2.2 Mean Residence Time

Name of the equation: Mean Residence Time

The equation:

$$\bar{t} = \int_0^\infty t\,E(t)\,dt = \frac{M_1}{M_0}$$

What the equation represents/how it's used: The average time a fluid element spends inside the reactor. For an ideal reactor with no dead zones, $\bar{t} = \tau = V/Q$.

Explain each term and the units of each term:

- $M_1 = \int_0^\infty t\,C(t)\,dt$: first moment of the tracer curve — units: mol $\cdot$ s^2/m^3
- $\tau = V/Q$: space time — V is reactor volume (m^3), Q is volumetric flow rate (m^3/s)

2.3 Variance of the RTD

Name of the equation: Variance of the RTD

The equation:

$$\sigma^2 = \int_0^\infty (t - \bar{t})^2\,E(t)\,dt = \frac{M_2}{M_0} - \bar{t}^2$$

What the equation represents/how it's used: Measures how spread out (wide) the RTD is. A PFR has $\sigma^2 = 0$ (no spread); a CSTR has $\sigma^2 = \tau^2$. More spreading means more deviation from ideal plug flow.

Explain each term and the units of each term:

- $M_2 = \int_0^\infty t^2\,C(t)\,dt$: second moment of the tracer curve — units: mol $\cdot$ s^3/m^3

2.4 Dimensionless Variance

Name of the equation: Dimensionless Variance

The equation:

$$\sigma_\theta^2 = \frac{\sigma^2}{\bar{t}^2}$$

What the equation represents/how it's used: A normalized measure of RTD spread used to diagnose reactor non-idealities and fit models. For an ideal PFR: $\sigma_\theta^2 = 0$. For an ideal CSTR: $\sigma_\theta^2 = 1$.

2.5 Dead Volume Diagnostic

Name of the equation: Dead Volume Diagnostic

The equation:

$$V_{dead} = V_{total}\left(1 - \frac{\bar{t}}{\tau}\right)$$

What the equation represents/how it's used: If the measured mean residence time $\bar{t}$ is less than the nominal space time τ, some fraction of the reactor volume is stagnant (dead) and not participating in the flow.

Explain each term and the units of each term:

- V_{total}: total physical reactor volume — units: m^3 or L
- $\bar{t}$: measured mean residence time — units: s
- τ: nominal space time $= V_{total}/Q$ — units: s

2.6 RTD of Ideal CSTR

Name of the equation: RTD of Ideal CSTR

The equation:

$$E(t) = \frac{1}{\tau}e^{-t/\tau}, \qquad t \geq 0$$

What the equation represents/how it's used: The RTD of a perfectly mixed tank (CSTR) is an exponential decay. There is a large probability of short residence times and a long exponential tail of long ones.

2.7 RTD of Ideal PFR

Name of the equation: RTD of Ideal PFR

The equation:

$$E(t) = \delta(t - \tau)$$

What the equation represents/how it's used: In a perfect plug flow reactor, all fluid elements spend exactly time τ in the reactor. The RTD is a Dirac delta function (spike) at $t = \tau$.

2.8 RTD of Two CSTRs in Series

Name of the equation: RTD of Two CSTRs in Series

The equation:

$$E(t) = \frac{1}{\tau_2 - \tau_1}\left(e^{-t/\tau_2} - e^{-t/\tau_1}\right)$$

What the equation represents/how it's used: The RTD of two CSTRs with different space times connected in series is the convolution of their individual RTDs. Crucially, this result is symmetric in τ_1 and τ_2 — the order does not matter for the RTD.

Explain each term and the units of each term:

- τ_1, τ_2: space times of each CSTR ($= V_i/Q$) — units: s

3. Tanks-in-Series (TIS) Model

3.1 TIS RTD

Name of the equation: TIS RTD

The equation:

$$E_n(t) = \frac{n}{\bar{t}} \cdot \frac{\left(\frac{nt}{\bar{t}}\right)^{n-1}}{(n-1)!}\,e^{-nt/\bar{t}}$$

What the equation represents/how it's used: The RTD of n equal-volume CSTRs in series. As $n \to \infty$, this approaches the PFR delta function. This is the tanks-in-series model used to represent non-ideal reactors with a unimodal RTD.

Explain each term and the units of each term:

- n: number of equivalent tanks (does NOT need to be an integer; can be fitted from σ_θ^2) — dimensionless
- $\bar{t}$: mean residence time — units: s

3.2 Number of Equivalent Tanks from Variance

Name of the equation: Number of Equivalent Tanks from Variance

The equation:

$$n = \frac{1}{\sigma_\theta^2}$$

What the equation represents/how it's used: Directly links the measured dimensionless variance

of the RTD to the number of equivalent CSTRs needed to model the reactor. A large σ_θ^2 means few tanks (well-mixed); a small σ_θ^2 means many tanks (approaching PFR).

4. Ideal Reactor Design Equations & Damköhler Number

4.1 Damköhler Number

Name of the equation: Damköhler Number

The equation (1st order):

$$Da = k\bar{t}$$

The equation (2nd order):

$$Da = kC_{A0}\bar{t}$$

What the equation represents/how it's used: The Damköhler number is the ratio of the characteristic reaction timescale to the characteristic residence time. $Da \gg 1$ means reaction is fast (high conversion); $Da \ll 1$ means reaction is slow (low conversion).

Explain each term and the units of each term:

- k: rate constant — units: s^{-1} (1st order) or $\mathrm{m}^3/(\mathrm{mol}\cdot\mathrm{s})$ (2nd order)
- $\bar{t}$: mean residence time — units: s
- C_{A0}: inlet concentration of A — units: $\mathrm{mol/m}^3$

4.2 Conversion in a Batch Reactor

Name of the equation: Conversion in a Batch Reactor

The equation (1st order):

$$X_{batch}(t) = 1 - e^{-kt}$$

The equation (2nd order):

$$X_{batch}(t) = 1 - \frac{1}{1 + kC_{A0}t}$$

What the equation represents/how it's used: Fractional conversion X of reactant A in a well-mixed batch reactor as a function of time t.

4.3 Conversion in a CSTR

Name of the equation: Conversion in a CSTR

The equation (1st order):

$$X_{CSTR} = 1 - \frac{1}{1 + Da}$$

The equation (2nd order):

$$X_{CSTR} = 1 - \frac{-1 + \sqrt{1 + 4Da}}{2Da}$$

What the equation represents/how it's used: Steady-state fractional conversion in a single ideal CSTR, derived from the CSTR design equation (mole balance on a perfectly mixed tank).

4.4 CSTR Design Equation (Quadratic Form for 2nd Order)

Name of the equation: CSTR Design Equation (Quadratic Form for 2nd Order)

The equation:

$$k\tau C_{out}^2 + C_{out} - C_{in} = 0$$

Solving for C_{out}:

$$C_{out} = \frac{-1 + \sqrt{1 + 4k\tau C_{in}}}{2k\tau}$$

What the equation represents/how it's used: The mole balance on a CSTR for a 2nd-order reaction. When solving two CSTRs in series, you apply this equation sequentially: the outlet of stage 1 becomes the inlet of stage 2.

Explain each term and the units of each term:

- $\tau = V/Q$: space time of the CSTR — units: s or min
- C_{in}: concentration entering the tank — units: mol/L
- C_{out}: concentration leaving the tank — units: mol/L

4.5 Conversion in a PFR

Name of the equation: Conversion in a PFR

The equation (1st order):

$$X_{PFR} = 1 - e^{-Da}$$

The equation (2nd order):

$$X_{PFR} = 1 - \frac{1}{1 + Da}$$

What the equation represents/how it's used: Fractional conversion in an ideal plug flow reactor, where every fluid element sees the same history (no axial mixing). For a given Da, PFR always achieves higher (or equal) conversion than a CSTR for $n > 1$ kinetics.

4.6 Conversion in Tanks-in-Series (TIS)

Name of the equation: Conversion in Tanks-in-Series (TIS)

The equation (1st order):

$$X_{TIS} = 1 - \left(1 + \frac{Da}{n}\right)^{-n}$$

What the equation represents/how it's used: Conversion for n equal CSTRs in series with 1st order kinetics, where Da is the overall Damköhler number based on total mean residence time. As $n \to \infty$, this recovers the PFR result.

5. Axial Dispersion (Dispersion Model)

5.1 1-D Dispersion Model (Governing Equation)

Name of the equation: 1-D Dispersion Model (Governing Equation)

The equation:

$$D_{ax}\frac{\partial^2 C}{\partial z^2} - \bar{u}\frac{\partial C}{\partial z} + R(C) = 0$$

What the equation represents/how it's used: The 1-D steady-state species balance in a tubular reactor that lumps radial nonuniformity (e.g., laminar flow profile) into a single axial dispersion coefficient D_{ax}. It is the workhorse model for quantifying deviation from ideal plug flow.

Explain each term and the units of each term:

- D_{ax}: axial dispersion coefficient — units: m^2/s
- $\bar{u}$: mean axial velocity — units: m/s
- z: axial coordinate — units: m
- $R(C)$: volumetric reaction rate (negative for consumption) — units: $\mathrm{mol}/(\mathrm{m}^3\cdot\mathrm{s})$

5.2 Péclet Number

Name of the equation: Péclet Number

The equation:

$$Pe = \frac{\bar{u}L}{D_{ax}} = \frac{\text{convection}}{\text{dispersion}}$$

What the equation represents/how it's used: The ratio of convective transport to dispersive (spreading) transport over the length of the reactor. $Pe \gg 1$: plug flow (little dispersion). $Pe \ll 1$: well-mixed (lots of dispersion).

Explain each term and the units of each term:

- $\bar{u}$: mean fluid velocity — units: m/s
- L: reactor length — units: m
- D_{ax}: axial dispersion coefficient — units: m^2/s

5.3 Wehner–Wilhelm Solution (1st Order, Dispersion Model)

Name of the equation: Wehner–Wilhelm Solution (1st Order, Dispersion Model)

The equation:

$$\frac{C_{out}}{C_{in}} = \frac{4q\, e^{Pe/2}}{(1+q)^2 e^{qPe/2} - (1-q)^2 e^{-qPe/2}}$$

where

$$q = \sqrt{1 + \frac{4Da}{Pe}}$$

What the equation represents/how it's used: The analytical solution for outlet/inlet concentration ratio in the 1-D dispersion model with Danckwerts boundary conditions and 1st-order kinetics. You need both Pe (how much dispersion) and Da (how much reaction) to use this.

Explain each term and the units of each term:

- $Pe = \bar{u}L/D_{ax}$: Péclet number — dimensionless
- $Da = kL/\bar{u}$: Damköhler number (for 1st order, dispersion model form) — dimensionless
- q: intermediate variable — dimensionless

5.4 Dispersion Model RTD Variance (Shortcut for $Pe > 10$)

Name of the equation: Dispersion Model RTD Variance (Shortcut for $Pe > 10$)

The equation:

$$\sigma_\theta^2 \approx \frac{2}{Pe}$$

What the equation represents/how it's used: For high Péclet numbers ($Pe > 10$), this shortcut links the measured dimensionless RTD variance to the Péclet number without solving the full Wehner–Wilhelm equation. You can measure σ_θ^2 from a tracer experiment and immediately get Pe.

5.5 Waste Multiplier

Name of the equation: Waste Multiplier

The equation:

$$\text{Waste Multiplier} = \frac{1 - X_{actual}}{1 - X_{PFR}}$$

What the equation represents/how it's used: Compares the unreacted reactant leaving a real reactor (with $\sigma_\theta^2 > 0$) to that from an ideal PFR at the same Da. A value > 1 means the real reactor wastes more reactant than an ideal PFR.

6. Taylor–Aris Dispersion (Dispersion from First Principles)

6.1 Taylor–Aris Axial Dispersion Coefficient

Name of the equation: Taylor–Aris Axial Dispersion Coefficient

The equation:

$$D_{ax} = \mathcal{D}_{AB} + \frac{\bar{u}^2 d_t^2}{192\mathcal{D}_{AB}} = \mathcal{D}_{AB} + \frac{\bar{u}^2 R_t^2}{48\mathcal{D}_{AB}}$$

Valid when $L >> \bar{u}R_t^2/\mathcal{D}_{AB}$

What the equation represents/how it's used: The effective axial dispersion coefficient in a straight tube with laminar flow, derived by Taylor and Aris. The first term is molecular diffusion alone; the second term (the "Taylor term") is the extra spreading caused by the laminar velocity profile coupling with diffusion.

Explain each term and the units of each term:

- $\mathcal{D}_{AB}$: molecular diffusivity of the solute in the fluid — units: m^2/s
- $\bar{u}$: mean axial fluid velocity — units: m/s
- d_t: tube inner diameter — units: m
- r_t: tube inner radius — units: m

6.2 Dimensionless Numbers (Fluid Mechanics)

Name of the equation: Dimensionless Numbers (Fluid Mechanics)

Schmidt Number:

$$Sc = \frac{\nu}{\mathcal{D}}$$

Reynolds Number:

$$Re = \frac{\bar{u}d_t}{\nu}$$

What the equation represents/how it's used: Sc is the ratio of momentum diffusivity (kinematic viscosity) to mass diffusivity. Re is the ratio of inertial to viscous forces — it determines whether flow is laminar or turbulent.

Explain each term and the units of each term:

- ν: kinematic viscosity of the fluid — units: m^2/s
- $\mathcal{D}$: molecular diffusivity — units: m^2/s
- $\bar{u}$: mean velocity — units: m/s
- d_t: tube diameter — units: m

7. Coil Reactor Dispersion (Taylor–Aris + Coiling)

7.1 Coil Reactor Dispersion Coefficient

Name of the equation: Coil Reactor Dispersion Coefficient

The equation:

$$\frac{1}{\kappa} = \frac{D_{ax,st}}{D_{ax,coil}} = 1 + \sqrt{Sc \cdot Ge}$$

therefore

$$D_{ax,coil} = \frac{D_{ax,st}}{1 + \sqrt{Sc \cdot Ge}}$$

What the equation represents/how it's used: Coiling a tube induces Dean flow (secondary flow

vortices), which enhances radial mixing and reduces axial dispersion relative to a straight tube. The larger the Germano number Ge, the more suppressed the axial dispersion.

7.2 Germano Number

Name of the equation: Germano Number

The equation:

$$Ge = \frac{Re^2 d_t}{2\rho_c}$$

What the equation represents/how it's used: A dimensionless number that quantifies the strength of the Dean vortices (secondary flow) induced by coiling. A larger Ge means stronger radial mixing and lower D_{ax}.

Explain each term and the units of each term:

- Re: Reynolds number — dimensionless
- d_t: tube diameter — units: m
- ρ_c: geometrical curvature radius of the coil — units: m

7.3 Geometrical Curvature Radius of the Coil

Name of the equation: Geometrical Curvature Radius of the Coil

The equation:

$$\rho_c = \sqrt{R_c^2 + \left(\frac{p}{2\pi}\right)^2} \approx R_c$$

What the equation represents/how it's used: The effective radius of curvature of a helical coil, accounting for both the coil radius R_c and the pitch p. For small pitches, $\rho_c \approx R_c$. Increasing the pitch p increases ρ_c, which decreases Ge, which increases D_{ax}.

Explain each term and the units of each term:

- R_c: coil radius (center of tube to center of helix) — units: m
- p: pitch of the coil (axial advance per full loop) — units: m

8. Reaction Networks & Theory of Reactor Design

8.1 Instantaneous Fractional Yield (Point Selectivity)

Name of the equation: Instantaneous Fractional Yield (Point Selectivity)

The equation:

$$\varphi = \frac{dC_R}{-dC_A} = \frac{r_R}{r_A}$$

What the equation represents/how it's used: The local ratio of the rate of desired product R formation to the rate of reactant A consumption at any point in the reactor. Optimizing how φ varies with C_A guides reactor choice.

Explain each term and the units of each term:

- C_R: concentration of desired product R — units: mol/m^3
- C_A: concentration of reactant A — units: mol/m^3
- r_R: rate of formation of R — units: mol/(m$^3 \cdot$ s)
- r_A: rate of consumption of A — units: mol/(m$^3 \cdot$ s)

8.2 Overall Yield

Name of the equation: Overall Yield

The equation:

$$\Phi = \frac{C_{Rf}}{C_{A0} - C_{Af}} = \bar{\varphi}$$

What the equation represents/how it's used: The overall yield is the moles of desired product formed divided by the moles of reactant consumed over the entire reactor. It equals the average of the instantaneous fractional yield φ.

Explain each term and the units of each term:

- C_{Rf}: exit concentration of desired product R — units: mol/m^3
- C_{A0}: inlet concentration of A — units: mol/m^3
- C_{Af}: exit concentration of A — units: mol/m^3

8.3 Theory of Reactor Design — Parallel Reaction Rules

Name of the equation: Theory of Reactor Design — Parallel Reaction Rules

For a parallel reaction network $A \to R$ (desired) and $A \to S$ (undesired):

$$r_R = k_1 C_A^{\alpha_1}, \qquad r_S = k_2 C_A^{\alpha_2}$$

$$\varphi = \frac{r_R}{r_R + r_S}$$

Rules:

- If $\alpha_1 > \alpha_2$: keep C_A high $\to$ use PFR (suppress mixing)
- If $\alpha_1 < \alpha_2$: keep C_A low $\to$ use CSTR (promote mixing)

What the equation represents/how it's used: The optimal reactor type for a parallel reaction network is determined entirely by the relative orders of the competing reactions with respect to the key reactant.

8.4 Van de Vusse Reaction Dimensionless Rate Constants

Name of the equation: Van de Vusse Reaction Dimensionless Rate Constants

For the van de Vusse network: $A \xrightarrow{k_1} Q \xrightarrow{k_2} S$ **and** $2A \xrightarrow{k_3} R$**:**

$$a_1 = \frac{k_3 C_{A0}}{k_1}, \qquad a_2 = \frac{k_2}{k_1}$$

What the equation represents/how it's used: Dimensionless rate constants that inform the relative natural rates of each reaction. a_1 compares the 2nd-order side reaction to the 1st-order main reaction; a_2 compares the downstream series step to the main reaction.

8.5 Fractional Yield Equation for CSTR vs. PFR

Name of the equation: Fractional Yield Equation for CSTR vs. PFR

CSTR:

$$\frac{x_Q}{x_A} = \frac{r_Q}{r_A}$$

PFR:

$$\frac{dx_Q}{dx_A} = \frac{r_Q}{r_A}$$

What the equation represents/how it's used: For a CSTR, the exit fractional yield equals the ratio of rates evaluated at exit conditions. For a PFR, it is a differential relationship that must be integrated. These equations are how you solve for selectivity vs. conversion in multi-reaction systems.

Explain each term and the units of each term:

- $x_A = (C_{A0} - C_A)/C_{A0}$: conversion of A — dimensionless
- $x_Q = C_Q/C_{A0}$: dimensionless concentration of desired product Q — dimensionless

9. Micromixing

9.1 Segregated Flow Model (Maximum Segregation)

Name of the equation: Segregated Flow Model (Maximum Segregation)

The equation:

$$\langle X \rangle = \int_0^\infty X_{batch}(t)\, E(t)\, dt$$

What the equation represents/how it's used: In the limit of maximum segregation, fluid elements never exchange material with each other. Each blob reacts like an isolated batch reactor for time t, and the overall conversion is the average weighted by the RTD. This is one bound on conversion for non-linear kinetics.

9.2 Mixing Time Estimate (Kolmogorov Microscale)

Name of the equation: Mixing Time Estimate (Kolmogorov Microscale)

The equation:

$$t_{mix} \sim \frac{L_{blob}^2}{\mathcal{D}_{AB}}$$

What the equation represents/how it's used: The time required for molecular diffusion to homogenize a "blob" of fluid of size L_{blob}. L_{blob} is taken from the Kolmogorov microscale (the smallest turbulent eddy size). If $t_{mix} \ll \bar{t}$, mixing is fast and maximum-mixedness applies; if $t_{mix} \gg \bar{t}$, segregated flow applies.

Explain each term and the units of each term:

- L_{blob}: characteristic blob (eddy) size from Kolmogorov microscale — units: m
- $\mathcal{D}_{AB}$: molecular diffusivity of A in B — units: m^2/s

10. External Mass Transfer (Film Resistance)

10.1 External Mass Transfer Flux

Name of the equation: External Mass Transfer Flux

The equation:

$$N_A = k_c(C_{Ab} - C_{As})$$

What the equation represents/how it's used: The molar flux of reactant A from the bulk fluid to the outer surface of the catalyst pellet across a thin diffusion "film." At steady state this equals the rate of reaction at the surface.

Explain each term and the units of each term:

- N_A: molar flux of A — units: mol/(m$^2 \cdot$ s)
- k_c: external mass transfer coefficient — units: m/s
- C_{Ab}: bulk concentration of A in the fluid — units: mol/m^3
- C_{As}: concentration of A at the pellet external surface — units: mol/m^3

10.2 Frossling (Ranz–Marshall) Correlation for k_c

Name of the equation: Frossling (Ranz–Marshall) Correlation for k_c

The equation:

$$Sh = \frac{k_c d_p}{\mathcal{D}} = 2 + 0.6 Re^{1/2} Sc^{1/3}$$

What the equation represents/how it's used: The Sherwood number Sh is the dimensionless mass transfer coefficient. This empirical correlation (Frossling/Ranz–Marshall) relates k_c to the fluid mechanics (Re) and fluid properties (Sc). At zero flow ($Re = 0$), $Sh = 2$ (pure diffusion around a sphere).

Explain each term and the units of each term:

- Sh: Sherwood number — dimensionless
- k_c: mass transfer coefficient — units: m/s
- d_p: pellet (particle) diameter — units: m
- $\mathcal{D}$: molecular diffusivity — units: m^2/s
- $Re = \bar{u} d_p/\nu$: Reynolds number based on particle diameter — dimensionless
- $Sc = \nu/\mathcal{D}$: Schmidt number — dimensionless

10.3 External Effectiveness Factor

Name of the equation: External Effectiveness Factor

The equation (general):

$$\eta_{ext} = \frac{\text{observed rate}}{\text{rate if } C_{As} = C_{Ab}}$$

At steady state (1st order reaction):

$$\eta_{ext} = \frac{1}{1 + Da_{ext}}, \qquad Da_{ext} = \frac{k\, C_{As}}{k_c\, a}$$

What the equation represents/how it's used: How much the external film resistance slows the observed rate relative to the intrinsic rate. $\eta_{ext} \to 1$: no external resistance. $\eta_{ext} \to 0$: strongly film-limited.

Diagnostic: If the observed Arrhenius activation energy $E_{obs} < 20$ kJ/mol, you are likely in the external mass transfer regime.

11. Internal (Pore) Diffusion & Thiele Modulus

11.1 Effective Diffusivity in a Porous Pellet

Name of the equation: Effective Diffusivity in a Porous Pellet

The equation:

$$D_{eff} = \frac{\varepsilon_p}{\tau_p} D_{pore}$$

where D_{pore} is either molecular or Knudsen diffusivity depending on pore size.

What the equation represents/how it's used: The effective diffusivity accounts for the fact that the interior of a pellet is not open space: the porosity ε_p reduces the available cross section, and the tortuosity τ_p lengthens the actual path a molecule must travel.

Explain each term and the units of each term:

- ε_p: pellet porosity (void fraction) — dimensionless, typically 0.3–0.6
- τ_p: tortuosity factor — dimensionless, typically 2–7
- D_{pore}: pore diffusivity (molecular or Knudsen) — units: m^2/s

Rule of thumb: $D_{eff} \sim 10^{-3}$ cm^2/s for gases and $\sim 10^{-6}$ cm^2/s for liquids.

11.2 Thiele Modulus (1st Order, Spherical Pellet)

Name of the equation: Thiele Modulus (1st Order, Spherical Pellet)

The equation:

$$\phi = R_p \sqrt{\frac{k}{D_{eff}}}$$

or using the Aris characteristic length $L_c = V_p/S_p = R_p/3$ for a sphere:

$$\phi = L_c \sqrt{\frac{k}{D_{eff}}}$$

What the equation represents/how it's used: The single most important parameter governing internal mass transfer limitations. It is the ratio of the characteristic reaction rate to the characteristic diffusion rate.

Limits:

- $\phi \ll 1$: fast diffusion, kinetic control — flat concentration profile, entire pellet reacts
- $\phi \gg 1$: fast reaction, diffusion control — steep gradient, pellet interior starved of reactant

Explain each term and the units of each term:

- R_p: pellet radius — units: m
- $L_c = V_p/S_p$: Aris characteristic length (volume-to-surface ratio) — units: m
- k: 1st-order intrinsic rate constant — units: s^{-1}
- D_{eff}: effective diffusivity inside pellet — units: m^2/s

11.3 Generalized Thiele Modulus (Any Reaction Order)

Name of the equation: Generalized Thiele Modulus (Any Reaction Order)

The equation:

$$\phi_{gen} = R_p \sqrt{\frac{k\, C_s^{\,n-1}(n+1)}{2D_{eff}}}$$

What the equation represents/how it's used: Extends the Thiele modulus concept to n-th order kinetics. For $n = 1$, it reduces to the standard Thiele modulus. For $n \neq 1$, it accounts for the concentration-dependence of the reaction rate in the pore.

Explain each term and the units of each term:

- C_s: concentration of reactant at the external surface of the pellet — units: mol/m^3
- n: reaction order — dimensionless
- All others defined above

11.4 Internal Effectiveness Factor

Name of the equation: Internal Effectiveness Factor

The equation (general):

$$\eta = \frac{\text{observed rate (with pore diffusion)}}{\text{rate if entire pellet were at surface concentration } C_s}$$

For 1st order, spherical pellet:

$$\eta = \frac{3}{\phi}\left(\frac{1}{\tanh(3\phi)} - \frac{1}{3\phi}\right)$$

Asymptotic limits:

- $\phi \ll 1$: $\eta \approx 1$
- $\phi \gg 1$: $\eta \approx \dfrac{1}{\phi}$ (or $\eta \sim 1/\phi_{gen}$ in the generalized case)

What the equation represents/how it's used: η tells you what fraction of the intrinsic pellet capacity you are actually using. In the diffusion-limited regime ($\phi \gg 1$), $\eta \approx 1/\phi$, meaning you waste most of the catalyst interior.

11.5 Membrane Reactor Thiele Modulus Scaling

Name of the equation: Membrane Reactor Thiele Modulus Scaling

The equation:

$$\Phi_M = \Phi_g \cdot \frac{2}{\left(\frac{R_2}{R_1}\right)^2 - 1}$$

where Φ_g is the generalized Thiele modulus using R_1 as the length scale.

What the equation represents/how it's used: The modified Thiele modulus for an annular membrane reactor (lumen + reactive shell). It corrects for the annular geometry, where R_1 is the inner (lumen) radius and R_2 is the outer radius. A smaller membrane thickness (smaller $R_2/R_1 - 1$) gives a smaller Φ_M, meaning less internal diffusion limitation and higher conversion.

Explain each term and the units of each term:

- Φ_g: generalized Thiele modulus (using R_1 as length scale) — dimensionless
- R_1: inner radius of the membrane (lumen outer wall) — units: m
- R_2: outer radius of the membrane — units: m

12. Diagnostics for Mass Transfer Limitations

12.1 Weisz–Prater Criterion

Name of the equation: Weisz–Prater Criterion

The equation:

$$N_{WP} = \frac{(-r_A)_{obs}\, \rho_c\, R_p^2}{D_{eff} C_{As}} \ll 1 \qquad \text{(no pore diffusion limitation)}$$

What the equation represents/how it's used: A dimensionless observable criterion — computed entirely from measurable quantities (no need to know the intrinsic k). $N_{WP} \ll 1$ confirms kinetic control; $N_{WP} \gg 1$ signals severe pore diffusion limitation.

Explain each term and the units of each term:

- $(-r_A)_{obs}$: measured (observed) rate of consumption of A per unit catalyst volume — units: $mol/(m^3 \cdot s)$
- ρ_c: catalyst pellet density — units: kg/m^3
- R_p: pellet radius — units: m
- D_{eff}: effective pore diffusivity — units: m^2/s
- C_{As}: concentration at external surface — units: mol/m^3

12.2 Falsified (Apparent) Activation Energy

Name of the equation: Falsified (Apparent) Activation Energy

The equation (strong diffusion control):

$$E_{obs} = \frac{E_{true}}{2}$$

What the equation represents/how it's used: In the diffusion-limited regime ($\phi \gg 1$), the observed Arrhenius slope halves the true activation energy because $\eta \sim 1/\phi \sim \sqrt{D_{eff}/k}$, and $k \sim e^{-E_{true}/RT}$. This halving of activation energy in an Arrhenius plot is a diagnostic signature of strong pore diffusion limitation.

12.3 Koros–Nowak (Madon–Boudart) Test

Name of the equation: Koros–Nowak (Madon–Boudart) Test

The equation:

$$s = \left.\frac{d\ln(-r_A)_{obs}}{d\ln(\rho_c)}\right|_{fixed\ d_p,\ C_{As}}$$

Rules:

- $s = 1$: kinetic control (rate proportional to active site density)
- $s = 0.5$: strong pore diffusion control ($\eta \sim 1/\phi \sim 1/\sqrt{k} \sim 1/\sqrt{\rho_c}$)

What the equation represents/how it's used: Rather than changing pellet size (which changes hydrodynamics), the Koros–Nowak test changes the density of active material inside a fixed-size pellet. It is the most rigorous test for distinguishing kinetic vs. diffusion control.

13. Selectivity in Parallel Reactions with Internal Mass Transfer

13.1 Observed Selectivity (Diffusion-Limited Regime)

Name of the equation: Observed Selectivity (Diffusion-Limited Regime)

For two parallel reactions $A \to R$ (desired, order n_1) and $A \to S$ (undesired, order n_2):

In kinetic control ($\phi \ll 1$):

$$\frac{r_R}{r_S} = \frac{k_1}{k_2} C_A^{\,n_1 - n_2}$$

In strong diffusion control ($\phi \gg 1$):

$$\frac{r_{R,obs}}{r_{S,obs}} = \frac{k_1}{k_2} C_A^{(n_1-n_2)/2} \cdot \sqrt{\frac{D_{eff,R}}{D_{eff,S}}}$$

What the equation represents/how it's used: Internal diffusion flattens the effective reaction order dependence: the exponent on C_A is halved compared to the kinetic regime. If $n_1 > n_2$ (desired reaction is higher order), diffusion limitation hurts selectivity to R.

14. Miscellaneous Models

14.1 Free-Zone Mixing (Feed-Point Mixing) Model

Name of the equation: Free-Zone Mixing (Feed-Point Mixing) Model

What the equation represents/how it's used: A multi-zone model for a reactor where reactants A and B are fed at different points, creating an A-rich zone and a mixing zone. At low recirculation ratio $\rho = Q_r/Q_2$, poor mixing keeps an A-rich zone where an undesired higher-order reaction in A dominates, lowering selectivity but increasing conversion.

14.2 Recycle Reactor Conversion (1st Order)

Name of the equation: Recycle Reactor Conversion (1st Order)

The equation:

$$\frac{C_{out}}{C_{in}} = \frac{1 + Re^{-Da/(1+R)}}{1+R}$$

Limits:

- $R = 0$: ideal PFR $\left(C_{out}/C_{in} = e^{-Da}\right)$
- $R \to \infty$: ideal CSTR $\left(C_{out}/C_{in} = 1/(1+Da)\right)$

What the equation represents/how it's used: The recycle reactor interpolates between PFR and CSTR behavior by recycling a fraction of the outlet stream back to the inlet. The recycle ratio $R = Q_{recycle}/Q_{feed}$ sets the degree of back-mixing.

Explain each term and the units of each term:

- R: recycle ratio (volumetric flow recycled / fresh feed flow) — dimensionless
- $Da = k\tau$: Damköhler number — dimensionless